

HIVE

Single-Window Campus Engagement Hub

Complete Project Documentation, Product Specifications,
Technical Architecture & Security Guidelines.

Target Institution: GH Rasoni College (GHR)

Prepared for: GHR College Management & Evaluators

Platform Components: Web App (Vite/React) & Backend Service (Node/Express)

Database: Supabase PostgreSQL (Postgres Cloud Server)

Document Version 1.0.0 (Production-ready Pitching Build)

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1. Product Requirements Document (PRD)

The Product Requirements Document (PRD) outlines the core business rationale, problems solved, and operational features of the HIVE platform designed specifically for GH Raisonni College (GHR).

1.1 Problem Statement

In modern educational campuses like GH Raisonni College, student communication and event organization are severely fragmented:

- **Communication Gaps:** Students miss crucial campus workshops, tech fests, and cultural events because notifications are scattered across WhatsApp groups, Telegram chats, and offline notice boards.
- **Manual Operations:** Club coordinators spend valuable hours manually registering students, tracking fests, verifying event attendance, and updating attendance registers.
- **Lack of Motivation:** There is no gamified reward system or central directory to encourage students to participate in co-curricular activities, making it difficult for campus clubs to sustain engagement.
- **Inefficient Query Resolution:** Students struggle to get real-time answers about event guidelines, points, schedules, or admin panels.

1.2 Target Users

- **Students (Primary Users):** Tech-savvy students seeking a single-window dashboard to discover fests, register, track points, and interact with peers.
- **Club Administrators:** Club coordinators, department heads, or student volunteers who organize events, scan tickets, verify check-ins, and export registers.
- **College Administrators:** Deans, HODs, or management staff who supervise all campus activities, approve club fests, manage user blocks, and review statistics.

1.3 Product Vision

To build HIVE - the ultimate single-window campus engagement hub for GH Raisonni College. HIVE gamifies campus life by tracking student participation through a centralized points leaderboard, automates event registration via custom QR-code ticket scans, provides a localized campus AI assistant ('Hey GHR') to answer student queries, and centralizes official announcements and chats.

1.4 Core MVP Features

- **Centralized Event Management:** Creation, deletion, and editing of campus events by verified Club and College Admins.
- **Mandatory Profile Details:** Mandatory fields for Name, Division, Year, and Department, ensuring robust student tracking.

- **QR-Code Ticket Check-in:** Mobile camera scanning of student QR tickets to automate event check-ins and verify attendance in real-time.
- **Campus AI Assistant (Hey GHR):** An intelligent conversational helper integrated with Google Gemini API to resolve campus, event, and study queries in Hinglish and English.
- **Leaderboards & Points:** A real-time points ranking list tracking student participation to gamify and encourage engagement.

2. Technical Architecture Document (TAD)

The Technical Architecture Document (TAD) details the technical stack, codebase structure, and hosting details governing the GHR HIVE application.

2.1 Technology Stack

- **Frontend Framework:** React 19, TypeScript, Vite (v7.1.3), Tailwind CSS, and Framer Motion.
- **Backend Engine:** Node.js with Express.js runtime, utilizing Server-Sent Events (SSE) for real-time chat broadcasts.
- **Database Management:** Supabase hosted PostgreSQL Cloud Server. Utilizes Transaction Pooler on Port 6543 to bypass Render's outbound IPv6 network constraints.
- **Hosting Infrastructure:** Vercel (Frontend SPA) & Render (Backend Web Service).

2.2 Directory Layout

The repository is cleanly split into frontend components and backend logic:

- proto1/vercel.json: Rewrites non-static URLs to index.html to fix SPA routing.
- proto1/backend/server.mjs: Houses API endpoints, SSE updates, and user authentication.
- proto1/backend/db.mjs: Configures the PostgreSQL client pool, migrations, and SQL schema queries.
- proto1/src/lib/api.ts: Coordinates fetch calls and frontend state logic.
- proto1/src/pages/HeyGHR.tsx: Incorporates the Web Speech API voice input and Gemini chatbot.

2.3 Database Tables

- **users:** id, email (Unique), name, password_hash, role (student/club_admin/Admin), points, div, year, department, blocked_at, avatar_url.
- **sessions:** token (Primary Key), user_id, expires_at.
- **events:** id, title, description, category, date, venue, organizer_id, capacity, points, image_url.
- **event_registrations:** event_id, user_id, registered_at, attended_at (tracks successful check-in timestamp).

3. Security & Access Document (SAD)

The Security & Access Document (SAD) details how user privacy is guarded, role-based checks are verified, and database transactions are kept clean.

3.1 Authentication & Restriction

- **Google OAuth Integration:** Login is powered by Google Identity Services (GSI) and whitelists college domains (.raisoni.net, ghrcem.edu), blocking arbitrary personal emails.

- **Session Security:** All private API calls require a database-backed bearer token, checked securely via session headers.

3.2 User Roles & Matrix

Permissions are strictly separated on both client and server-side:

1. Student: Read posts/events, register, view own points, chat with HeyGHR.
2. Club Admin: Add events, scan QR tickets to log student attendance, export registrations.
3. College Admin: Supervise all database tables, review club certificates, manage user states (block/unblock), edit/delete any content.

3.3 Error Handling & Edge Cases

- **Query Constraints:** SQL schemas specify target columns in all conflict statements (e.g. followers or registration uniques) to prevent query failures.

- **DoS Safety:** Incoming payload body limits (MAX_BODY_BYTES = 5MB) prevent RAM overflow exploits.

- **Gemini API Fallback:** If the AI API key is missing or fails, the chatbot smoothly switches to regex local rules so the site remains functional.

4. Frontend Specification Document (FSD)

The Frontend Specification Document (FSD) details the design system, animations, UI components, and client integration rules.

4.1 Color Palette & Styling

- **Primary Brand Purple:** #7C3AED - represents campus innovation and youth.
- **Accent Colors:** Indigo (#4F46E5) and soft violet-to-pink gradient panels.
- **Backgrounds:** #F5F3FF (Soft Light Purple) transitioning to #EDE9FE.
- **Bubble Glassmorphic Design:** Assistant bubbles use `rgba(255,255,255,0.95)` with `backdrop-filter: blur(10px)` and a soft card shadow.

4.2 Spacing & Responsive Rules

- Desktop Screen Layout: Dual sticky sidebar system. Left navigation sidebar, center feed column, and right-hand side column (stories/recent fests).
- Mobile Screen Layout: Locked to max-width 450px centered shell to simulate a native app look, utilizing a bottom navigation bar.

4.3 Micro-Animations & Integrations

- **Web Speech API Integration:** Clicking the chatbot mic icon triggers HTML5 SpeechRecognition. Transcribes mixed Hindi/English speech directly into the text input box.
- **HTML5 Camera Scanner:** Uses the browser's video stream in real-time to decode check-in tickets inside the Club/Admin panel.
- **Excel CSV Generation:** Outputs instant tabular packages of event registration records directly to the coordinator's download folder.

5. Feature Ticket List (FTL)

The Feature Ticket List details the status of all core technical components deployed for the HIVE application:

Ticket ID	Feature / Task	Status
T-01	Supabase PostgreSQL Database Migration	COMPLETED
T-02	Mandatory Student Profile Field Validation	COMPLETED
T-03	Google OAuth restriction to college email domains	COMPLETED
T-04	Gemini 2.5 Flash API Assistant integration (HeyGHR)	COMPLETED
T-05	Web Speech API mic integration for voice typing	COMPLETED
T-06	Vercel SPA Routing and rewrite rules (vercel.json)	COMPLETED
T-07	QR Attendance in-app camera scanner integration	COMPLETED
T-08	Attendance Live Excel CSV Exporter Utility	COMPLETED